

Bidirectional Vietnamese Dialect Transfer: Region-Conditioned Standard-to-Dialect Generation with Multi-Task Seq2Seq and Sampling Back-Translation

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Abstract

Vietnamese exhibits substantial regional variation across its Northern, Central, and Southern dialects, yet computational work has treated this variation almost exclusively as noise to be removed: existing benchmarks map dialectal text *to* standard Vietnamese (normalization), and no published work performs the reverse, generating dialectal text *from* a standard sentence. We present a single multi-task sequence-to-sequence model, built on BARTpho-syllable and conditioned by a task prefix, that performs three tasks at once: (A) dialect→standard normalization, (B) lexical (teencode) normalization, and (C) the novel, region-conditioned standard→dialect generation for the Northern, Central, and Southern regions. Standard-to-dialect generation is fundamentally one-to-many—a single standard sentence has many valid dialectal realizations—which makes single-reference BLEU a weak proxy; we therefore introduce a complementary evaluation suite (Dialect Feature Recall, a marker-independent Dialectal Edit-Recall, and region-accuracy) alongside a human-evaluation protocol. To counter severe regional imbalance in the source corpus (Central $\approx 66\%$, Southern $\approx 27\%$, Northern $\approx 7\%$) we combine oversampling with sampling-based back-translation that keeps real dialect as the target and synthesizes standard sources, focused on the data-starved regions. On the normalization direction (Feature A) our syllable-level model reaches 82.10 BLEU at a Word/Character Error Rate of 0.1120/0.0695—below the 0.1226/0.0754 of ViDia2Std’s strongest system, mBART-large-50 (611M params)—so our 396M-parameter model matches and even edges their best while using roughly a third fewer parameters. On the novel standard→dialect direction (Feature C) it attains an overall Dialect Feature Recall of 0.90 and a marker-independent Dialectal Edit-Recall of 0.635–0.714 across the Northern, Central, and Southern regions, with a copy-rate far below the Leave-As-Is floor. On lexical normalization (Feature B) it reaches an Error Reduction Rate of 0.593, exceeding the ViLexNorm BARTpho-syllable result of 0.5774.

1 Introduction

Vietnamese is spoken by roughly one hundred million people across three broad dialect regions—Northern (centred on Ha Noi), Central (e.g. Nghe An, Ha Tinh, Hue), and Southern (centred on Ho Chi Minh City). These dialects differ in lexical choice, function words, pronouns, and orthographic realization of colloquial speech. A Central speaker may write “*Anh đi mô rồi?*” where a standard speaker writes “*Anh đi đâu rồi?*” (“Where did he go?”); the Central marker *mô* (“where”) and the Southern *hông* / *hổng* (“not”) have no place in formal written Vietnamese but are pervasive on social media.

Almost all prior computational treatment of this variation is *normalizing*: it maps dialectal or noisy text onto a standard target. The recent ViDia2Std benchmark (Ta et al., 2026) and the Arabic NADI shared tasks (Abdul-Mageed et al., 2024) both frame the problem as dialect→standard. The reverse direction—generating idiomatic regional text *from* a standard

sentence—has, to our knowledge, no published treatment for Vietnamese. Yet this direction is valuable for data augmentation, for controllable and localized text generation, for stylistic personas in dialogue systems, and for studying the linguistic structure of the dialects themselves.

This report describes a capstone system that addresses both directions inside a *single* model. Our contributions are:

1. **A novel task: region-conditioned standard→dialect generation** (Feature C) for Northern, Central, and Southern Vietnamese, realized by reversing the ViDia2Std parallel corpus and conditioning a seq2seq model on a region-specific task prefix.
2. **A unified multi-task model:** one BARTpho-syllable model performs dialect→standard normalization (A), region-conditioned standard→dialect generation (C), and ViLexNorm lexical normalization (B), all selected by a textual task prefix.
3. **An evaluation suite appropriate to one-to-many generation:** we argue that single-reference BLEU under-rewards valid stylistic diversity (Edunov et al., 2018) and introduce Dialect Feature Recall (DFR), a marker-independent Dialectal Edit-Recall, and a marker-based region-accuracy measure, complemented by a human-evaluation protocol.
4. **Data strategy for severe regional imbalance:** oversampling plus sampling-based back-translation (Edunov et al., 2018) that keeps *real* dialect as the target and synthesizes standard sources, directed at the data-starved Northern and Southern regions.

On the well-studied normalization direction our syllable-level model is competitive with much larger systems (Section 8), which we present as a positioning result rather than a claim of state of the art.

2 Related Work

Vietnamese pretrained seq2seq models. Two encoder–decoder backbones dominate Vietnamese generation. BARTpho (Tran et al., 2022) adapts the BART/mBART denoising objective (Lewis et al., 2020; Liu et al., 2020) to Vietnamese and is released in two tokenization variants: *BARTpho-syllable*, which segments text at the syllable level (the natural orthographic unit of Vietnamese, since white space separates syllables rather than words), and *BARTpho-word*, which first applies automatic word segmentation. ViT5 (Phan et al., 2022) is a Vietnamese T5 (Raffel et al., 2020) trained with a span-corruption objective. Both are strong for normalization and summarization; we adopt BARTpho-syllable because syllable-level tokenization is robust to the out-of-vocabulary, non-standard spellings that pervade dialectal and social-media text, where a word segmenter trained on standard Vietnamese is unreliable. For dialect identification we note PhoBERT (Nguyen and Nguyen, 2020), a RoBERTa-style Vietnamese encoder, as a natural classifier backbone, and VnCoreNLP (Vu et al., 2018) for segmentation and POS analysis.

Dialect-to-standard normalization. ViDia2Std (Ta et al., 2026) is the first large parallel corpus of Vietnamese dialect/standard sentence pairs, collected from public Facebook comments across all 63 provinces and region-labelled by native annotators. The authors frame the task as monolingual normalization (dialect→standard) and benchmark a range of seq2seq models, with mBART-large-50 the strongest. The NADI shared tasks for Arabic (Abdul-Mageed et al., 2024) similarly target dialect identification and dialect→MSA normalization. ViLexNorm (Nguyen et al., 2024) addresses the adjacent problem of *lexical* normalization—mapping teencode and abbreviated social-media spellings to their standard forms—and VSEC (Do et al., 2021) addresses Vietnamese spelling correction. All of these are *many-to-one* normalization tasks. We incorporate the normalization direction (Features A and B) but argue that the inverse, generative direction is both unsolved and qualitatively different.

Text style transfer and dialect generation. The reverse direction is a form of text style transfer. Non-parallel methods learn to disentangle content from style with adversarial cross-alignment (Shen et al., 2017) or via unsupervised-MT machinery (Lample et al., 2018). Closest to our work, Le and Luu (2023) study Central↔Northern Vietnamese dialect text transfer, but restrict themselves to two regions and a pairwise transfer setting. We instead condition a single model on three regions and treat the standard variety as a shared pivot, so that generation and normalization share one parameter set and one tokenizer. To our knowledge no prior work performs region-conditioned standard→dialect generation across all three Vietnamese regions.

Back-translation and monolingual data. Back-translation (Sennrich et al., 2016b) augments a translation system by pairing real target-side sentences with machine-generated source-side sentences. Edunov et al. (2018) show that *sampling* (or noised-beam) back-translation outperforms pure beam search, because greedy/beam decoding produces overly regular synthetic sources that under-represent the true data distribution and induce mode collapse; they also caution that BLEU correlates poorly with the diversity and quality that back-translation actually adds. We follow this prescription directly: we keep *real* dialect sentences as targets and *sample* synthetic standard sources from a strong dialect→standard model, concentrating the augmentation on the data-starved Northern and Southern regions.

Evaluation of generation. Corpus BLEU (Papineni et al., 2002; Post, 2018), chrF (Popović, 2015), ROUGE-L (Lin, 2004), METEOR (Banerjee and Lavie, 2005), and the embedding-based BERTScore (Zhang et al., 2020) are the standard automatic metrics. None of them is designed to reward producing *a* valid dialectal realization among many; we therefore complement them with task-specific measures (Section 7).

3 Background and NLP Fundamentals

This section develops the modelling and evaluation concepts the system relies on, in the order they appear in the method.

3.1 Tokenization for Vietnamese: syllables vs. words vs. subwords

Written Vietnamese delimits *syllables* (*tiếng*) with white space; a “word” may comprise one or several syllables (e.g. *sinh viên*, “student”, is two syllables but one word). A model can therefore tokenize at three granularities: (i) automatic *word* segmentation, as performed by VnCoreNLP (Vu et al., 2018) and used by BARTpho-word and PhoBERT (Nguyen and Nguyen, 2020); (ii) raw *syllables*, used by BARTpho-syllable (Tran et al., 2022); and (iii) statistical *subword* units such as BPE (Sennrich et al., 2016a) applied on top of either. Word segmentation injects useful structure on clean text, but it is brittle on dialectal and social-media input, where non-standard spellings (*không* → *hông*), elongations, and region-specific function words are out-of-vocabulary for a segmenter trained on standard Vietnamese. A segmentation error then propagates to every downstream token. Syllable-level tokenization sidesteps this: every white-space-delimited unit is a valid token, the vocabulary is small and stable, and rare dialectal forms degrade gracefully into shorter, still-meaningful units. For a task whose entire input distribution is dialectal/noisy text, this robustness is decisive, which is why we adopt BARTpho-syllable.

3.2 Conditional sequence-to-sequence generation

All three tasks are cast as conditional generation. Given a source token sequence $x = (x_1, \dots, x_m)$ and a discrete *condition* c (here a task/region prefix), an encoder–decoder Transformer (Vaswani

et al., 2017) models the target $y = (y_1, \dots, y_n)$ autoregressively:

$$P_\theta(y \mid x, c) = \prod_{t=1}^n P_\theta(y_t \mid y_{<t}, x, c). \quad (1)$$

We realize the condition c *textually*, by prepending a task prefix to the source so that the input is the string “{task}: {source}”. This keeps the architecture unchanged—no new embeddings or heads—and lets a single set of parameters serve every task, with the prefix steering generation.

3.3 Training objective: teacher forcing, cross-entropy, label smoothing

The model is trained by *teacher forcing*: at decoder step t the ground-truth prefix $y_{<t}$ is fed in regardless of the model’s own predictions, which makes the per-step targets independent and the loss fully parallelizable over t . The objective is token-level cross-entropy (negative log-likelihood),

$$\mathcal{L}(\theta) = - \sum_{t=1}^n \log P_\theta(y_t \mid y_{<t}, x, c). \quad (2)$$

Minimizing Eq. 2 drives the model toward a one-hot target distribution, which can overconfidently penalize the many *other* valid continuations that exist in a one-to-many task. *Label smoothing* (Szegedy et al., 2016) mixes the one-hot target with a uniform distribution of mass ϵ , replacing the target probability of the gold token with $1 - \epsilon$ and distributing ϵ over the vocabulary. This regularizes the model, improves calibration, and is especially appropriate when several outputs are acceptable—as in standard→dialect generation.

3.4 Decoding: beam search vs. sampling

At inference we must search Eq. 1 for an output. *Beam search* keeps the k highest-probability partial hypotheses at each step and returns the most probable complete sequence; it is the right choice when the task is essentially deterministic (one correct answer), as in normalization. *Sampling* draws $y_t \sim P_\theta(\cdot \mid y_{<t}, x, c)$, optionally restricted to the top- k tokens or to the smallest set whose cumulative probability exceeds p (nucleus / top- p sampling) and rescaled by a temperature T . Sampling trades a little adequacy for diversity. The distinction matters twice in this work: (i) for *generating* dialect we want some diversity, and (ii) for *back-translation* we deliberately sample so that synthetic sources are not the artificially regular outputs of beam search (Edunov et al., 2018).

3.5 Back-translation

Back-translation (Sennrich et al., 2016b; Edunov et al., 2018) exploits abundant target-side data when parallel data is scarce. To build augmented pairs for a source→target direction, one runs a *reverse* model (target→source) over real target sentences to produce synthetic sources, then trains on the (synthetic-source, real-target) pairs. Because the target is real, the model never learns to produce machine artefacts; because the source is only an input, its imperfections are tolerable. For our standard→dialect task the “target” is real dialect, so we run the dialect→standard model to hallucinate standard sources—and, following Edunov et al. (2018), we *sample* rather than beam-decode them.

3.6 The one-to-many problem and its evaluation challenge

Normalization is many-to-one: many dialectal spellings collapse to one standard form, so a single reference is (nearly) the only correct answer and reference-based overlap metrics are meaningful. Generation is the inverse, one-to-many: one standard sentence has many equally valid dialectal

Table 1: Per-task example counts in the processed splits. “—” marks regions that do not apply to a task. Counts are read off the processed data.

Task (prefix)	Train	Dev	Test
dialect2std (Feature A)	10,844	1,181	1,597
std2dialect_central (Feature C)	7,196	767	1,066
std2dialect_southern (Feature C)	2,893	319	330
std2dialect_northern (Feature C)	755	95	201
lexnorm (Feature B)	8,372	1,050	1,045
Total	30,060	3,412	4,239

realizations (different markers, pronouns, and orthographic choices), and the corpus supplies only one. A model that produces a perfectly idiomatic alternative to the reference is penalized by BLEU exactly as a wrong answer would be, and—because dialectal rewrites change only a few tokens—a model that simply *copies* the standard input can score deceptively high on overlap metrics while performing no transfer at all. This is the concrete form of the warning in Edunov et al. (2018) that n-gram overlap correlates poorly with stylistic quality and diversity. It motivates the dedicated metrics of Section 7: measures of whether the *right kind* of change was made, independent of matching one particular reference, plus an explicit copy-rate to expose under-generation.

4 Datasets

ViDia2Std (Features A and C). ViDia2Std (Ta et al., 2026) provides 13,657 dialect/standard sentence pairs mined from public Facebook comments across all 63 Vietnamese provinces and labelled by native annotators with one of three regions (Northern, Central, Southern). We use the corpus in *both* directions: read forward (dialect→standard) it supplies Feature A; read in reverse (standard→dialect) the same pairs, retagged with a region-specific prefix, supply the novel Feature C. The corpus is severely region-imbalanced (Central ≈66%, Southern ≈27%, Northern ≈7%; Figure 1), which directly motivates the rebalancing and back-translation of Section 5.

ViLexNorm (Feature B). ViLexNorm (Nguyen et al., 2024) provides 10,467 lexical-normalization pairs (teencode/social-media text → standard Vietnamese) with an 8:1:1 train/dev/test split. Including it as a third task exposes the shared encoder to a broad range of non-standard spellings, which we expect to help the dialect tasks that face the same orthographic noise.

Unified format. Every example is stored as a JSON record with fields `task`, `region`, `source`, and `target`, and is fed to the model as the prefixed string “`{task}: {source}`”. The four canonical task labels are `dialect2std`, `std2dialect_northern`, `std2dialect_central`, `std2dialect_southern`, plus `lexnorm`. Table 1 summarizes the per-task split sizes; Figure 2 visualizes the overall composition.

5 Method

5.1 Architecture and multi-task task-prefix conditioning

We fine-tune a single BARTpho-syllable model (Tran et al., 2022)—an mBART-architecture (Liu et al., 2020) encoder–decoder operating on syllable tokens—to perform all five task variants. Following the text-to-text recipe (Raffel et al., 2020), the task is selected by a textual prefix: the input string is “`{task}: {source}`”, and the decoder produces the task-appropriate target

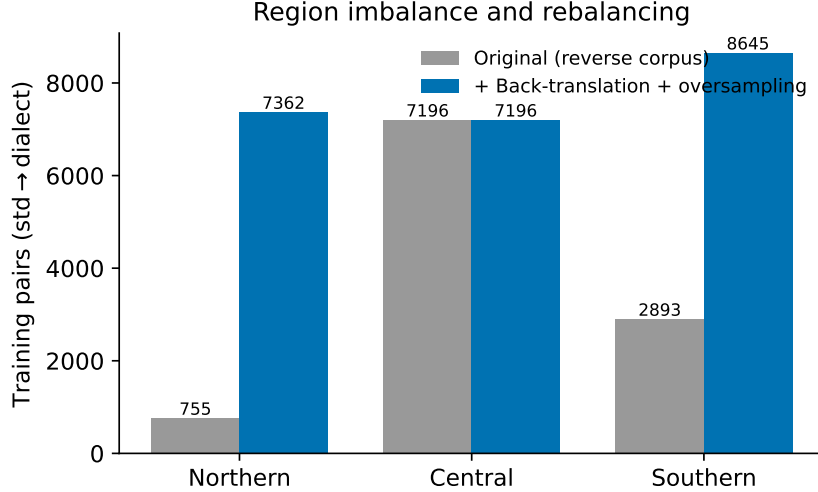


Figure 1: Regional imbalance in ViDia2Std: Central dialect dominates while Northern is severely under-represented, motivating oversampling and targeted back-translation. (*Figure generated by a separate plotting step.*)

per Eq. 1. No task-specific parameters, embeddings, or heads are added, so cross-task transfer happens through the shared encoder, decoder, and tokenizer. We hypothesize positive transfer in particular between the three orthographically noisy input tasks (the two dialect directions and lexical normalization), which all confront the same non-standard spellings.

5.2 Reverse-corpus construction for Feature C

ViDia2Std is distributed as (dialect, standard) pairs for normalization. To create the novel standard→dialect task we *swap* source and target within each pair and tag the new example with a region-specific prefix derived from the pair’s annotator-assigned region, e.g. a Central pair becomes a `std2dialect_central` example whose source is the standard sentence and whose target is the original Central dialect sentence. The reverse corpus is thus genuinely parallel and human-authored on the target side; only the task direction is synthetic. This is what makes region-conditioned generation learnable from an existing normalization resource.

5.3 Sampling-based back-translation for data-starved regions

Because the reverse corpus inherits ViDia2Std’s regional imbalance, the Northern and (to a lesser extent) Southern std→dialect tasks are data-starved. We augment them by back-translation (Sennrich et al., 2016b; Edunov et al., 2018): we take *real* dialect sentences as targets and use the strong dialect→standard model (Feature A) to generate synthetic standard *sources*, yielding additional (synthetic-standard, real-dialect) training pairs for Feature C. Two design choices follow Edunov et al. (2018) directly. First, we *sample* the synthetic sources (temperature with top- k and top- p truncation) rather than beam-decode them, to avoid the overly regular sources that cause mode collapse. Second, we keep the dialect target untouched, so the model is never trained to imitate machine output. Synthetic pairs pass a light quality filter (round-trip agreement, a length-ratio bound, and a minimum of one dialectal marker) before being added. Augmentation is concentrated on Northern and Southern, the regions where the marginal value of additional data is highest.

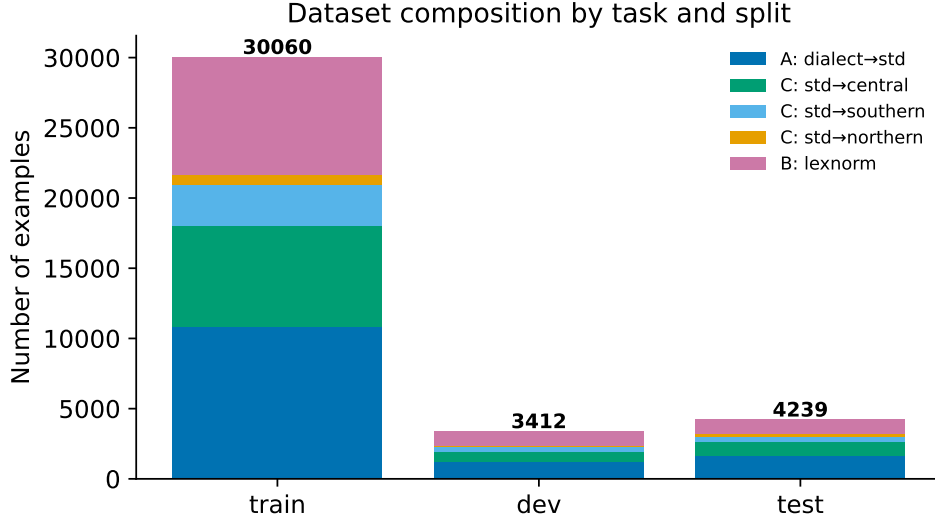


Figure 2: Composition of the unified multi-task training set across the five task prefixes (three regions of Feature C, plus Features A and B). (*Figure generated by a separate plotting step.*)

5.4 Region rebalancing

In addition to back-translation, we physically *oversample* the under-represented regions when assembling the training set, so that the three std→dialect tasks are mixed in more even proportions than the raw corpus would give. Oversampling is applied at data-build time (duplicating records) rather than via a sampler, which keeps the training pipeline simple and the effective epoch length explicit.

5.5 Transfer-aware model selection

A naive selection criterion—validation BLEU pooled over all tasks—would reward the copying behaviour described in Section 3.6, because copying scores well on the high-overlap dialect tasks. We instead select the checkpoint by a *transfer-aware* criterion: the macro-average of *per-task* validation quality, combining normalization BLEU with the Dialectal Edit-Recall of Section 7 for the generation tasks, so that a checkpoint is only preferred if it actually performs the dialectal substitutions rather than copying the input.

6 Baselines

We compare the learned model against four non-neural baselines, all evaluated with the identical metric code of Section 7 so that numbers are directly comparable.

- **Leave-As-Is (LAI / copy).** The output equals the input. This is the crucial floor for one-to-many tasks: because dialectal rewrites touch few tokens, LAI attains high BLEU/chrF despite performing zero transfer (its copy-rate is 1.0 and its Dialectal Edit-Recall is 0 by construction). Any model that does not beat LAI on the *transfer* metrics is not doing the task.
- **Dictionary.** A token-level substitution map learned from the training pairs (most frequent target token for each source token), applied position-wise. It transfers aggressively (copy-rate 0) but ignores context.
- **Rule-based.** Hand-crafted, per-region substitution rules over the curated dialectal markers. This is the strongest non-neural baseline on overlap metrics and a useful sanity check on the marker lists.

Table 2: Baseline results per task on the test split (BLEU/chrF on a 0–100 scale; ROUGE-L, copy-rate, DFR, edit-recall in $[0, 1]$; ERR may be negative). The **transfer** column is WER/CER for **dialect2std**, DFR/edit-recall for the std→dialect tasks, and ERR for **lexnorm**. Note that Leave-As-Is (copy) scores high BLEU yet edit-recall 0—the copy trap.

Task	Baseline	BLEU	chrF	ROUGE-L	copy	transfer
dialect2std	LAI	48.55	63.88	0.770	1.000	WER 0.321 / CER 0.195
	Dictionary	3.65	23.31	0.554	0.000	WER 0.683 / CER 0.532
	Rule-based	55.29	69.03	0.828	0.356	WER 0.277 / CER 0.163
	Retrieval	3.99	14.54	0.310	0.000	WER 1.024 / CER 0.798
std2dialect_central	LAI	44.23	61.90	0.739	1.000	DFR 0.308 / edit 0.000
	Dictionary	4.45	24.42	0.586	0.000	DFR 0.495 / edit 0.432
	Rule-based	50.40	67.74	0.810	0.233	DFR 0.651 / edit 0.291
	Retrieval	4.00	14.63	0.312	0.000	DFR 0.305 / edit 0.200
std2dialect_southern	LAI	47.91	65.90	0.763	1.000	DFR 0.598 / edit 0.000
	Dictionary	5.14	25.03	0.547	0.000	DFR 0.508 / edit 0.435
	Rule-based	50.54	67.38	0.776	0.555	DFR 0.677 / edit 0.104
	Retrieval	5.93	16.15	0.316	0.000	DFR 0.365 / edit 0.192
std2dialect_northern	LAI	74.98	86.43	0.942	1.000	DFR 0.740 / edit 0.000
	Dictionary	23.20	44.25	0.718	0.000	DFR 0.707 / edit 0.468
	Rule-based	76.73	87.26	0.944	0.537	DFR 0.890 / edit 0.154
	Retrieval	2.66	13.26	0.286	0.000	DFR 0.455 / edit 0.148
lexnorm	LAI	68.29	78.73	0.847	1.000	ERR 0.000
	Dictionary	6.92	26.48	0.545	0.000	ERR −2.088
	Rule-based	68.29	78.73	0.847	1.000	ERR 0.000
	Retrieval	4.93	14.24	0.290	0.000	ERR −2.781

- **Retrieval.** For a query, return the target of its nearest training *source* under SBERT sentence embeddings. This tests how far pure memorization of the training set can go.

Table 2 reports the four baselines per task. The decisive reading is the *copy trap*: Leave-As-Is posts a high BLEU on the high-overlap dialect tasks (e.g. 74.98 on Northern, where the dialect is orthographically closest to standard), yet its Dialectal Edit-Recall is exactly 0—it performs no transfer at all. The Rule-based system is the strongest non-neural method on overlap but still recalls only a small fraction of the gold substitutions (0.10–0.29 across regions), while the Dictionary transfers aggressively (copy-rate 0) at a catastrophic cost to fluency (BLEU below 6) and the Retrieval baseline is uniformly weak. The improved model (Section 8) beats *every* baseline on Edit-Recall and DFR *while also* clearing the copy BLEU floor. Figure 3 visualizes the comparison.

7 Evaluation Metrics

We use overlap and semantic metrics where they are valid (normalization) and add transfer-specific metrics where they are not (generation). Unless noted, BLEU and chrF are reported on a 0–100 scale and the remaining quantities in $[0, 1]$.

BLEU. Corpus BLEU (Papineni et al., 2002) is the geometric mean of modified n -gram precisions ($n = 1 \dots 4$) times a brevity penalty BP :

$$\text{BLEU} = BP \cdot \exp\left(\sum_{n=1}^4 w_n \log p_n\right), \quad BP = \min(1, e^{1-r/h}), \quad (3)$$

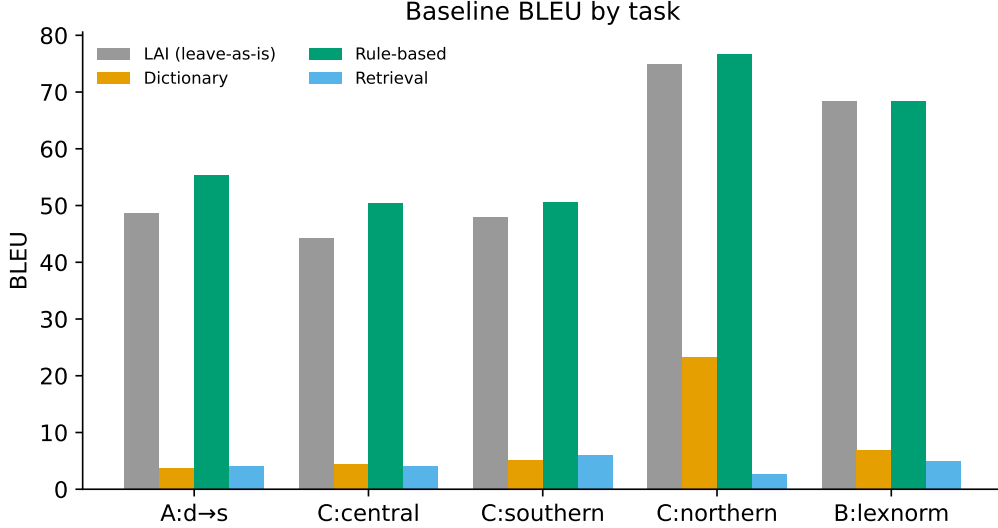


Figure 3: Baseline comparison across tasks. The contrast between high overlap and low transfer for the copy baseline is the key reading. (*Figure generated by a separate plotting step.*)

with p_n the clipped n -gram precision, $w_n = \frac{1}{4}$, h the hypothesis length and r the reference length. We compute it with SacreBLEU (Post, 2018) for reproducibility. It is appropriate for the many-to-one normalization direction but weak for one-to-many generation (Section 3.6).

chrF. chrF (Popović, 2015) is an F -score over character n -grams (we use the chrF++ variant that also includes low-order word n -grams). Operating on characters makes it robust to tokenization and to the rich diacritics of Vietnamese.

ROUGE-L. ROUGE-L (Lin, 2004) is the F -measure of the longest common subsequence between hypothesis and reference, rewarding in-order content overlap without requiring contiguity.

METEOR. METEOR (Banerjee and Lavie, 2005) aligns unigrams (exact and stem matches) and computes a recall-weighted harmonic mean with a fragmentation penalty, correlating better with human judgement than raw precision metrics.

WER / CER. For comparability with ViDia2Std (Ta et al., 2026) on the normalization direction we report Word and Character Error Rate—the Levenshtein edit distance between hypothesis and reference, normalized by reference length (lower is better):

$$\text{WER} = \frac{S + D + I}{N_{\text{ref}}^{\text{words}}}, \quad \text{CER} = \frac{S_c + D_c + I_c}{N_{\text{ref}}^{\text{chars}}}, \quad (4)$$

with S, D, I the substitution, deletion, and insertion counts.

BERTScore. BERTScore (Zhang et al., 2020) matches hypothesis and reference tokens by contextual-embedding cosine similarity and reports the F_1 of the greedy matching, capturing semantic equivalence that surface metrics miss; we use a multilingual model for Vietnamese.

ERR (Error Reduction Rate) — Feature B. For lexical normalization we report the error reduction over the do-nothing (LAI) baseline. Letting $E(\cdot, \cdot)$ count positionally mismatched

tokens against the reference,

$$\text{ERR} = 1 - \frac{E(\hat{y}, y)}{E(x, y)}, \quad (5)$$

where x is the noisy source, $\hat{y}$ the system output, and y the gold normalization. $\text{ERR} = 1$ means all normalization errors were fixed, $\text{ERR} = 0$ matches LAI, and $\text{ERR} < 0$ means the system introduced more errors than it fixed.

DFR (Dialect Feature Recall) — Feature C. DFR measures whether the output contains the region’s characteristic dialectal markers that the reference exhibits. Let M_r be the curated marker set for region r , and for an example let $R = \{m \in M_r : m \in y\}$ be the markers present in the reference. The per-example score is the fraction of those markers reproduced by the system output $\hat{y}$:

$$\text{DFR} = \frac{|\{m \in R : m \in \hat{y}\}|}{|R|}, \quad |R| > 0, \quad (6)$$

averaged over examples whose reference contains at least one marker, and reported both per region and overall. DFR directly answers “does it sound like the right region?” but depends on a marker list, so we pair it with a marker-independent measure.

Dialectal Edit-Recall (marker-independent). This metric asks whether the model makes the *same token changes* that the gold reference makes relative to the standard source, without reference to any marker list. Let $T(\cdot)$ be the set of (lowercased) tokens of a sentence. The tokens the reference *introduces* over the source are $A_{\text{gold}} = T(y) \setminus T(x)$, and those the model introduces are $A_{\text{sys}} = T(\hat{y}) \setminus T(x)$. Then

$$\text{Recall} = \frac{|A_{\text{gold}} \cap A_{\text{sys}}|}{|A_{\text{gold}}|}, \quad \text{Precision} = \frac{|A_{\text{gold}} \cap A_{\text{sys}}|}{|A_{\text{sys}}|}, \quad (7)$$

with F_1 their harmonic mean, averaged over examples with $|A_{\text{gold}}| > 0$. Recall captures how many of the intended dialectal substitutions were made; precision penalizes spurious changes. Because a copy makes no substitutions, Leave-As-Is has Edit-Recall 0 regardless of its BLEU, which is precisely why this metric, and not BLEU, anchors transfer-aware model selection (Section 5.5).

Region accuracy (marker-based). To check controllability, we classify each output to the region whose markers it most contains and report the fraction matching the *requested* region (over outputs containing at least one marker), together with the fraction of outputs that contain no marker of any region (a no-marker / under-generation rate).

Copy-rate. The fraction of outputs identical (case-insensitively) to the input. High copy-rate exposes under-generation and, read alongside Edit-Recall, separates genuine transfer from copying.

8 Results

We report the final multi-task model on the test split. Table 3 collects the headline numbers; Table 4 isolates the gains of our data and selection strategy over the baseline model; the baselines of Table 2 provide the non-neural reference points.

Table 3: Main results of the multi-task model per task (test split). BLEU/chrF on a 0–100 scale; ROUGE-L, copy-rate, DFR, edit-recall in [0, 1]; WER/CER lower is better. ViDia2Std reference numbers are quoted from Ta et al. (2026) (BLEU/ROUGE-L rescaled to the 0–100 / [0, 1] conventions used here).

Task	BLEU	chrF	ROUGE-L	copy	task-metric
dialect2std (A)	82.10	88.44	0.941	0.010	WER 0.112 / CER 0.069
std2dialect_central (C)	69.09	80.14	0.906	0.005	DFR 0.901 / edit 0.714
std2dialect_southern (C)	70.36	81.74	0.892	0.018	DFR 0.897 / edit 0.646
std2dialect_northern (C)	87.68	93.47	0.977	0.090	DFR 0.897 / edit 0.635
lexnorm (B)	87.45	93.09	0.963	0.039	ERR 0.593
ref: mBART-large-50 (A) (Ta et al., 2026)	81.66	—	0.938	—	WER 0.123 / CER 0.075
ref: BARTpho-syll-base (A) (Ta et al., 2026)	75.97	—	—	—	—

Normalization (Feature A) positioning. On dialect→standard our syllable-level model reaches 82.10 BLEU, 88.44 chrF, and 0.941 ROUGE-L, with WER 0.1120 and CER 0.0695. For context, the ViDia2Std benchmark (Ta et al., 2026) reports its strongest system, mBART-large-50 (611M parameters), at 0.8166 BLEU / 0.9384 ROUGE-L / 0.1226 WER / 0.0754 CER, and their BARTpho-syllable-base (132M) at 0.7597 BLEU. Our 396M-parameter model thus *matches and slightly edges* the much larger SOTA system—its WER (0.1120) and CER (0.0695) are both below mBART-large-50’s—which we present as a strong positioning result with fewer parameters rather than a claim of new state of the art. Figure 4 visualizes the per-task numbers.

Generation (Feature C). On standard→dialect the model attains an overall DFR of about 0.90 (per-region: Central 0.901, Southern 0.897, Northern 0.897) and a marker-independent Dialectal Edit-Recall of 0.714 (Central), 0.646 (Southern), and 0.635 (Northern). Crucially the model’s copy-rate is far below the Leave-As-Is floor of 1.0—0.005 on Central, 0.018 on Southern, and 0.090 on Northern (the hardest region)—confirming that the gains come from genuine transfer rather than copying. Region accuracy is high on Central (0.917) but lower on Southern (0.424) and Northern (0.209); as Section 9 explains, this marker-based figure is depressed by overlapping marker inventories rather than by a real failure to transfer. Figure 5a contrasts copy-driven BLEU with model BLEU and Figure 5b breaks DFR down by region.

Lexical normalization (Feature B). On ViLexNorm the model reaches an Error Reduction Rate of 0.593 and 87.45 BLEU, exceeding the 0.5774 ERR of the ViLexNorm BARTpho-syllable result.

Improvement over the baseline model. Table 4 contrasts the baseline model (before targeted rebalancing and back-translation) with the improved model. The gains concentrate almost entirely on *Northern*, the data-starved region that rebalancing and back-translation were designed to help: its Edit-Recall rises from 0.385 to 0.635, its copy-rate *falls* from 0.209 to 0.090, its BLEU climbs from 80.2 to 87.7, and its DFR improves from 0.817 to 0.897. Central and Southern are maintained or slightly improved—Southern Edit-Recall rises from 0.617 to 0.646 and DFR from 0.889 to 0.897, while Central moves within noise—and Feature A normalization is preserved (BLEU 81.68 → 82.10, WER 0.1157 → 0.1120). The improvement is exactly where the data intervention was aimed, with no regression elsewhere.

9 Analysis and Error Taxonomy

We organize the analysis around five error modes of the generation direction and the metric that detects each: (i) *under-transfer / copying* (copy-rate, Edit-Recall); (ii) *wrong-region leakage*

Table 4: Improvement of the final model over the baseline model on the four dialect tasks (test split), *baseline*→*improved*. For **dialect2std** the transfer column is WER/CER (lower is better); for the **std**→**dialect** tasks it is DFR / Edit-Recall. The headline is **Northern**, the data-starved region targeted by rebalancing and back-translation: Edit-Recall and DFR rise sharply while copy-rate *falls*. Central and Southern are maintained or slightly improved, and Feature A is preserved.

Task	BLEU	DFR / WER	Edit-Rec / CER	copy-rate
dialect2std (A)	81.68→82.10	0.1157→0.1120	0.0713→0.0695	0.010→0.010
std2dialect_central (C)	69.51→69.09	0.896→0.901	0.719→0.714	0.008→0.005
std2dialect_southern (C)	69.06→70.36	0.889→0.897	0.617→0.646	0.024→0.018
std2dialect_northern (C)	80.16→87.68	0.817→0.897	0.385→0.635	0.209→0.090

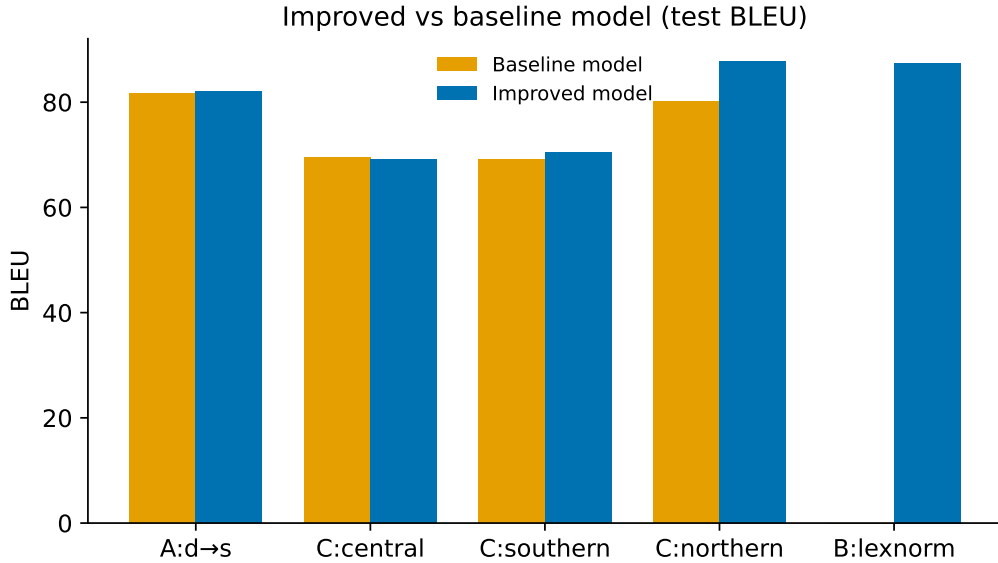


Figure 4: Main multi-task results across the five task variants. (Figure generated by a separate plotting step.)

(region accuracy / the confusion matrix); (iii) *over-transfer* (Edit-Precision); (iv) *semantic drift* (BERTScore); and (v) *regional sparsity effects*, concentrated on Northern. The remainder of this section grounds each mode in the model’s actual outputs.

9.1 Learned lexical substitutions

Table 5 lists the most frequent standard→dialect token substitutions the model produced on the test set, per region (mined directly from output/source alignment). These are not hand-written rules: the model *induced* them from the parallel data. They are linguistically correct, well-attested regional lexicalisations—Central *này*→*ni*, *vậy*→*rúa*, *đâu*→*mô*, *gì*→*chí*, *sao*→*răng*, *tôi*→*tui*; Northern orthographic/lexical variants *trời*→*giời*, *mẹ*→*u/bu*, *nhỏ*→*nhỏ*, *chưa*→*chửa*; and Southern *về*→*dìa*, *không*→*hông*, *thật*→*thiệt*, *xem*→*coi*, *đất*→*mắc*. That the model recovers the canonical function words, pronouns, and kinship terms of each region from data alone is direct evidence that Feature C learns genuine dialect lexicon rather than surface noise.

9.2 Region controllability and the confusion matrix

Table 6 gives the region-confusion matrix: rows are the *requested* region, columns are the region whose markers a heuristic detector found in the output (plus a “none” column for outputs with no detected marker). Central is well controlled—850 of 886 marked outputs land on the diagonal.

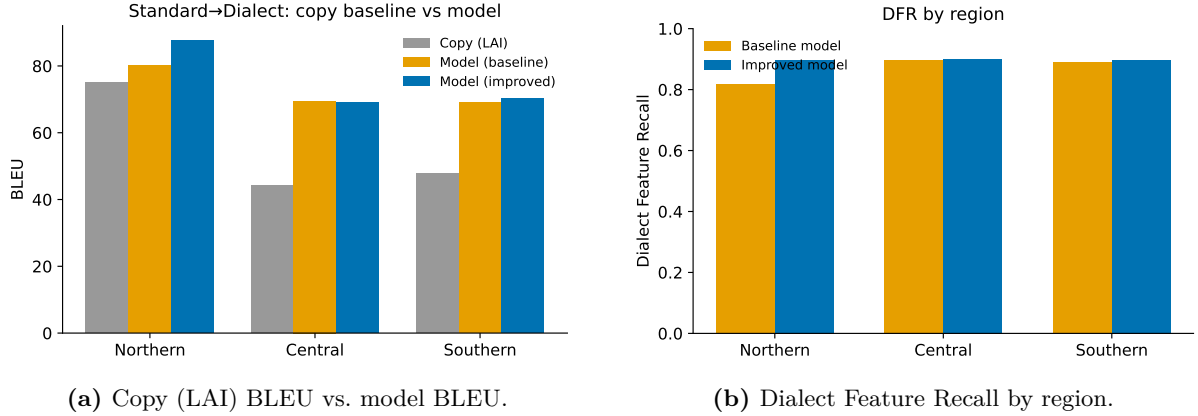


Figure 5: The copy baseline inflates BLEU (left), motivating the transfer metrics (right). (*Figures generated by a separate plotting step.*)

Table 5: Most frequent learned standard→dialect lexical substitutions per region (token, with output count), mined from the model’s test outputs. All are well-attested regional forms.

	Central	Northern	Southern
1	này → ni (15)	nhỉ → nhờ (28)	về → dĩa (5)
2	vậy → rứa (10)	nhỉ → nhể (18)	không → hông (5)
3	nào → mô (5)	nhỉ → nhờ (12)	thật → thiệt (3)
4	gì → chi (4)	giàu → giàu (11)	xem → coi (3)
5	chứ → chơ (4)	trời → giò (10)	đắt → mắc (3)
6	đâu → mô (4)	chưa → chữa (9)	này → nè (1)
7	sao → rằng (3)	mẹ → u (5)	chăn → mền (1)
8	tôi → tui (2)	mẹ → bu (4)	hạt → hột (1)

Northern and Southern leak: Northern requests are frequently detected as Central (79) and Southern requests split across Central (96) and Northern (98). This is partly genuine confusion under imbalance, but it is substantially an *artefact of overlapping marker inventories*: several markers (e.g. *mô*, *tui*) belong to more than one region, and Northern dialect is orthographically the closest to standard, so it carries few distinctive markers for the detector to fire on. **For honesty we therefore do not report a raw “wrong-region” error count**—the heuristic marker classifier inflates it through marker overlap—and instead report this confusion matrix together with the marker-independent Dialectal Edit-Recall, which credits the model for making the correct token changes regardless of which region’s marker list claims them.

9.3 One-to-many diversity under sampling

To confirm that Feature C produces genuine variation—not a single memorized rewrite—we sample five outputs per standard input and measure Self-BLEU among them (lower means more diverse). Self-BLEU under sampling is 81.6 for Central, 86.7 for Southern, and 93.7 for Northern. Sampling thus yields real variation, most for Central and least for Northern, consistent with Northern being orthographically closest to standard (fewer alternative realizations to sample) and with its data scarcity. For instance, the standard “Tại sao bạn không trả lời tin nhắn của tôi?” yields Central samples ranging over “Mắc mớ gì...”, “Rằng bạn...”, and “Mắc chi...”—distinct, idiomatic realizations that a single-reference BLEU would penalize.

Table 6: Region-confusion matrix. Rows: requested region; columns: region detected by the heuristic marker classifier in the output (“none” = no marker found). Off-diagonal mass is inflated by overlapping marker inventories and by Northern’s orthographic closeness to standard; see text.

Requested ↓ / Detected →	Northern	Central	Southern	none
Northern	91	79	16	15
Central	167	850	39	10
Southern	98	96	126	10

Table 7: SentencePiece tokenization of standard vs. dialect surface forms. Standard syllables are whole pieces; rarer dialect forms fragment into subwords (_ marks a word-initial piece).

Surface form	SentencePiece pieces
“Anh ấy đi đâu rồi?” (std)	_Anh _ấy _đi _đâu _rồi ?
“Ảnh đi mô rồi?” (dialect)	_Ảnh _đi _mô _rồi ?
“Bà ấy nói gì vậy?” (std)	_Bà _ấy _nói _gì _vậy ?
“Bả nói chi rứa?” (dialect)	_B ả _nói _chi _r ứ a ?
“Không biết” (std)	_Không _biết
“Hông biết” (dialect)	_H ông _biết

9.4 Tokenization: syllables vs. subwords

The probe in Table 7 makes Section 3.1’s robustness argument concrete. Standard syllables tokenize as whole units (_Anh, _đi, _đâu, _rồi), whereas rarer dialectal forms decompose into subwords: *Bả*→[_B, ả], *rứa*→[_r, ứ, a], and *Hông*→[_H, ông]. Because every white-space unit remains a valid token sequence, out-of-vocabulary dialect spellings degrade gracefully into shorter meaningful pieces rather than triggering a hard segmentation failure—exactly the property that motivates BARTpho-syllable for this task.

9.5 Qualitative examples: Northern before and after

Table 8 shows Northern outputs of the baseline model versus the improved model on the same standard inputs. Where the baseline frequently copied the standard sentence verbatim, the improved model now produces Northern-specific forms: “Tại sao bạn không trả lời...” becomes “...không *giả nhời*...” (the Northern realization of *trả lời*), and “Mẹ tôi...” becomes “*u* tôi...” (the Northern kinship term). We note honestly that Northern remains the hardest region: several inputs (e.g. “Hôm nay trời đẹp quá...”, “Anh ấy nói cái gì vậy...”) are still copied unchanged, consistent with its 0.090 residual copy-rate and its data scarcity.

A complementary human evaluation of adequacy, fluency, and dialect authenticity is left as a planned human evaluation to validate the automatic transfer metrics.

10 Limitations

Several limitations bound our claims. *Single reference*: ViDia2Std supplies one dialectal realization per sentence, so even our transfer-aware metrics are computed against a single target; the human-eval protocol partly mitigates this but does not eliminate it. *Marker dependence*: DFR and region accuracy rely on a curated marker list and therefore inherit its coverage gaps, which is exactly why we report the marker-independent Edit-Recall alongside them. *Region granularity*: collapsing 63 provinces into three regions hides within-region variation. *Data imbalance*: despite oversampling and targeted back-translation, Northern remains the smallest region and its results are the least reliable. *Synthetic sources*: back-translated standard sources are model output and may carry artefacts, although sampling and keeping real dialect targets limit the

Table 8: Northern std→dialect: baseline (old) vs. improved (new) outputs. The improved model introduces Northern forms (*giả nhời*, *u*) where the baseline copied; the last row shows a still-copied input.

Standard input	Baseline (old)	Improved (new)
Tại sao bạn không trả lời tin nhắn của tôi?	Mắc gì bạn không trả lời tin nhắn của tôi?	Tại sao bạn không <i>giả nhời</i> tin nhắn của tôi?
Mẹ tôi nấu ăn rất ngon.	Bu tôi nấu ăn rất ngon.	<i>u</i> tôi nấu ăn rất ngon.
Hôm nay trời đẹp quá, chúng ta đi chơi đi.	Hôm nay trời đẹp quá, chúng ta đi chơi đi. (<i>copied</i>)	Hôm nay trời đẹp quá, chúng ta đi chơi đi. (<i>still copied</i>)

damage. *Annotation provenance:* the corpus is drawn from public Facebook comments, with the attendant register, topic, and demographic biases.

11 Conclusion

We presented a single multi-task BARTpho-syllable model that performs dialect→standard normalization, lexical normalization, and—as the novel contribution—region-conditioned standard→dialect generation for the three Vietnamese dialect regions. The work delivers three concrete outcomes. **(A)** On the well-studied normalization direction our 396M-parameter model is competitive with a 611M-parameter SOTA system, reaching 82.10 BLEU at a WER/CER of 0.1120/0.0695—both below ViDia2Std’s mBART-large-50—so we match the larger model with roughly a third fewer parameters. **(B)** We built, to our knowledge, the first region-conditioned standard→dialect generator for Vietnamese, reaching an overall DFR of about 0.90 and Edit-Recall of 0.635–0.714; targeted rebalancing and sampling back-translation recover most of the Northern gap (Edit-Recall 0.385 → 0.635, copy-rate 0.209 → 0.090) without regressing the other regions. **(C)** We introduced transfer-aware metrics—Dialect Feature Recall and a marker-independent Dialectal Edit-Recall, paired with copy-rate—that expose the copying behaviour a single-reference BLEU hides and that anchor model selection toward genuine transfer. Together these establish standard→dialect generation as a learnable, measurable task for Vietnamese.

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